Transport Policy, Vol. 2, No. 3, pp. 153–158, 1995

0967-070X/95 $10.00 + 0.00

The demand for rail transport in European countries

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New data from fourteen European countries on passenger and freight rail transport reveal the significant role of fares, subsidies and, especially, quality variables in explaining the large variation in the demand for rail across Europe. High train frequencies in Switzerland and the Netherlands have contributed to a strong rail performance in the passenger market, and for Austria in the freight market. In Britain, high fares and low subsidies have led to a comparatively poor rail performance. Travel and haulage have shifted from rail to road resulting in external costs which are widely underestimated.

Keywords: rail transport, fares, subsidies, (frequency) freight

Introduction

In most Western European countries the demand for passenger rail travel is growing. However, given the much more rapid increase in the ownership and use of the private car, rail travel has witnessed a relative decline during the 1980s. The data in Table 1 confirm this generalization except in the cases of Switzerland, Eire and the Netherlands where the shares of rail in total travel have remained stable.[1] The picture is more mixed in the case of freight transport. Total freight tonne km transported in the fourteen country sample as a whole has declined slightly, though eight countries have in fact experienced marginal increases in freight traffic between 1980 and 1990. The story is much more uniform in terms of freight rail market share which has fallen in all but Austria and Portugal. Figure 1 plots changes in rail's passenger share against changes in its freight market share between 1980 and 1990.

From a public policy perspective, the displacement of rail by road transport matters because of the external costs generated by the latter. While good estimates are unavailable for Britain,[2] several recent studies converge to find external costs accounting for about 10–15% of GNP in the USA and Germany.[3] In view of these externalities that are very negligible for rail transport, present British Government policy of basing rail investment on purely commercial, pecuniary cost–benefit considerations seems flawed, foregoing as it does the enormous welfare gains available from a transfer from road to rail transport. Conventional cost function estimates for rail operations are thus not a satisfactory basis for policy decisions.

Differential external costs by mode also support arguments in favour of using fiscal policy to provide incentives for travellers to make socially optimal modal choices. Currently car owners face a private marginal cost of the trip which is very low compared to the sunk cost of car purchase and compared to the substantial external costs which road transport generates. In contrast, if a train is not full to capacity the marginal social cost of an extra passenger is effectively zero. But given that the fixed costs of a rail system are large, the policy of imposing some form of average cost or break-even pricing on rail transport pushes fares up far above the marginal social cost of travel. To provide a level playing field and permit car owners to make socially efficient modal decisions requires public subsidies for rail to cover the fixed or sunk costs of investment and also the imposition of additional taxes on road users. Table 1 shows that there is considerable variation between countries in the use of rail and road transport. The modal

The authors are very grateful to Professor C A Nash, Director of the Institute for Transport Studies, University of Leeds, for providing us with unpublished data on European rail systems and helpful suggestions. We retain responsibility for the results and opinions expressed.

[1] The share data refer to land based mechanized modes and therefore exclude air travel as well as cycling and trips on foot.

[2] In view of more comprehensive estimates for other countries, Pearce's (1993) finding of external costs amounting to only 3–4% of British GDP are almost certainly much too low.

[3] See Lowe (1994), Miller and Moffet (1993), Mackenzie *et al.* (1993) and Teufel (1991). These estimates are actually very conservative. They neglect the strong negative relationship between various health measures and proximity to traffic flows found by Whitelegg *et al.* (1993) and also the estimated 10 000 annual deaths from small particulates (PM10) largely produced by diesel emissions, reported in the *New Scientist* 12 March 1994.

Table 1 The demand for passenger and freight rail transport

		Passenger transport Km (bn)	% Share	Freight transport Tonne km (bn)	% Share
United Kingdom (BR)	1980	30.26	6.7	17.64	14.8
	1990	34.20	5.4	15.80	9.9
Switzerland (CFF)	1980	9.18	10.7	7.39	50.3
	1990	11.06	10.8	8.30	41.6
Eire (CIE)	1980	1.03	3.5	0.62	11.0
	1990	1.29	3.6	0.59	10.3
Portugal (CP)	1980	6.08	11.1	1.00	7.81
	1990	5.66	7.0	1.59	12.71
West Germany (DB)	1980	40.50	7.0	63.80	25.2
	1990	43.60	6.3	61.40	20.6
Denmark (DSB)	1980	4.31	8.7	1.62	17.1
	1990	4.86	7.1	1.79	16.0
Italy (FS)	1980	39.59	9.4	18.38	12.2
	1990	46.43	7.1	21.22	10.1
Netherlands (NS)	1980	8.89	6.9	3.40	5.7
	1990	11.06	6.9	3.07	4.6
Norway (NSB)	1980	2.75	7.1	1.66	24.0
	1990	2.43	5.1	1.63	14.3
Austria (OBB)	1980	7.38	11.6	11.00	39.9
	1990	8.46	11.1	12.68	46.1
Spain (RENFE)	1980	14.83	8.5	11.30	10.9
	1990	16.73	7.6	11.61	7.0
Belgium (SNCB)	1980	6.96	8.5	8.04	23.6
	1990	6.54	N/A	8.35	17.8
France (SNCF)	1980	54.66	10.0	66.37	31.6
	1990	63.74	9.2	51.53	26.7
Sweden (SJ)	1980	7.00	8.6	16.65	43.8
	1990	6.17	6.1	19.61	42.5
Total	1980	226.46	8.4	228.87	21.9
	1990	255.69	7.2	219.17	17.1

Source: Preston *et al.* (1994), table 2.1 and table 2.2.

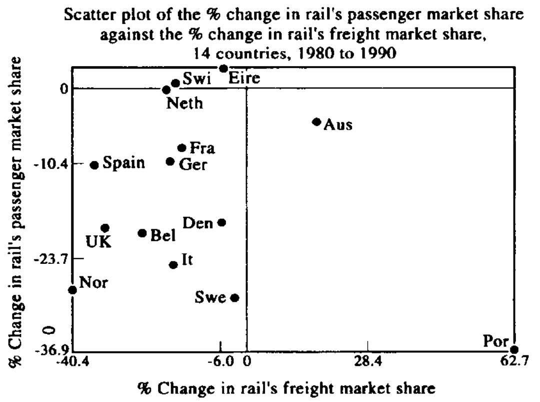

Figure 1

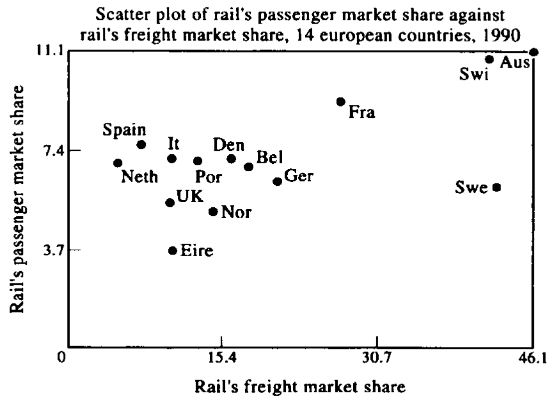

Figure 2

share for passenger travel in Switzerland and Austria, for example, is twice as favourable to rail as that in the UK. The inter-country variation in modal shares is even greater across countries for freight rail. It ranges from a maximum of 46.1% in Austria to a sample minimum of 4.6% in the Netherlands in 1990. Figure 2 plots a scatter diagram of rail's passenger market share in 1990 against its share of the freight market. There is a strong, positive association between the two shares; the simple correlation coefficient is 0.63.[4] In particular, Austria, Switzerland and, to a lesser extent, France stand out as strong performers in both the passenger and freight rail sectors.

[4] This positive association between rail's share in the freight and passenger markets does not hold for all developed countries. Japan, for example, has a high passenger share (31%) but a low freight share (5%). The USA is the converse with figures of 0.4% and 37% respectively (see Lowe, 1994, Table 2).

Table 2.1 Data common to both passenger and freight rail

	Population	Route	Real GNP
United Kingdom	233.2	0.068	5826
Switzerland	159.9	0.070	7835
Eire	51.4	0.028	3957
Portugal	110.3	0.037	2898
West Germany	246.6	0.108	6840
Denmark	119.1	0.054	6220
Italy	190.7	0.053	5932
Netherlands	433.6	0.082	5860
Norway	13.0	0.012	5893
Austria	90.6	0.063	6085
Spain	77.4	0.025	4370
Belgium	327.9	0.115	6166
France	102.7	0.062	6466
Sweden	18.9	0.028	6257
Mean	155.4	0.058	5758
Standard deviation	120.5	0.031	1245
Coefficient of variation	0.78	0.53	0.22

The source for these data is Preston *et al.* (1994). Population density is defined as population per square kilometre; route density is defined as rail route-km per square km. Real GNP per capita is converted into £ using PPP exchange rates. All data refer to 1990.

Table 2.2 Data specific to passenger rail

	Fare	Demand	Frequency	Petrol price	Station spacing
United Kingdom	6.21	599.3	17587	0.68	6.7
Switzerland	3.74	1634.1	23468	0.79	3.7
Eire	3.58	364.4	4951	0.94	16.1
Portugal	1.92	604.2	10179	0.88	4.1
West Germany	3.95	710.9	9298	0.74	6.1
Denmark	3.97	947.6	17359	0.97	8.3
Italy	2.37	808.3	13339	1.14	5.8
Netherlands	3.52	751.6	32829	0.87	7.2
Norway	3.78	578.6	4084	1.02	6.9
Austria	2.85	1042.9	8393	0.91	3.9
Spain	2.79	427.4	7217	0.72	7.7
Belgium	2.88	659.3	19406	0.83	6.2
France	3.38	1134.6	7972	0.89	6.9
Sweden	5.17	844.1	2260	1.09	19.8
Mean	3.58	793.4	12739	0.89	7.81
Standard deviation	1.10	325.4	8523	0.14	4.57
Coefficient of variation	0.31	0.41	0.67	0.15	0.58

All the data are drawn from Preston *et al.* (1994) except for petrol prices. The source for the petrol price data is the Department of Transport. It refers to the price of a litre of leaded petrol expressed in pence and converted to a common currency using purchasing parity exchange rates. Passenger fares are likewise measured in pence as passenger revenue per passenger-km. Demand is defined as passenger-km per capita. Frequency is proxied by total passenger train-km divided by route length. Station spacing refers to the average distance between passenger stations measured in km.

Table 2.3 Data specific to freight rail

	Fare	Demand	Frequency	Diesel price	Station spacing
United Kingdom	4.15	276.9	8467.9	0.66	132.7
Switzerland	4.09	1226.3	17631.5	0.78	5.0
Eire	3.72	166.7	2373.0	0.87	16.1
Portugal	2.58	169.7	2854.2	0.57	15.6
West Germany	4.51	1001.1	13106.9	0.63	9.7
Denmark	5.29	349.0	6162.8	0.71	8.3
Italy	2.37	368.4	6220.1	0.74	7.8
Netherlands	2.07	208.6	9098.2	0.57	39.4
Norway	2.75	388.1	4991.8	0.38	6.9
Austria	3.70	1563.1	12433.7	0.63	7.1
Spain	2.93	296.6	6477.7	0.54	10.4
Belgium	2.52	841.7	24799.6	0.61	8.4
France	3.38	917.2	6340.7	0.63	8.6
Sweden	1.23	2682.8	6964.9	0.67	13.1
Mean	3.24	747.0	9137.3	0.64	20.7
Standard deviation	1.08	710.3	6101.2	0.12	33.4
Coefficient of variation	0.34	0.95	0.67	0.18	1.62

All the data are drawn from Preston *et al.* (1994) except for diesel prices. The source for the diesel price data is the Department of Transport. It refers to the price of a litre of diesel expressed in pence and converted to a common currency using purchasing parity exchange rates. Freight price rates are likewise measured in pence as freight revenue per tonne-km. Demand is defined as freight tonne-km per capita. Frequency is proxied by total freight train-km divided by route length. Station spacing refers to the average distance between freight stations measured in km.

The aim of this paper it to explain these cross-country variations in the demand for passenger and freight rail travel. We use a new data set collected by the Institute of Transport Studies at the University of Leeds for a single survey year, 1990 (Preston *et al.*, 1994). Given the absence of data from previous years on the complete set of variables, we are unable to estimate any pooled cross-section-time-series models. The plan of this paper is to present in the next section the basic summary statistics for the data set. In the following section we specify a simple demand for rail function in terms of available cost components, including various measures of service quality which are novel in international comparative studies. In the subsequent section we present and discuss the OLS regression results. Conclusions and policy lessons are summarized in the closing section.

The data

Thirteen variables are used in the empirical analysis, three are common to both passenger and freight rail demand equations, five are specific to passenger travel and five are specific to freight. These data are listed in Tables 2.1, 2.2, and 2.3 respectively together with some descriptive summary statistics by variable for the fourteen country sample.

The basic geographical variable used to compare countries of very different size and population is simply population density. Kilometres of rail route per square kilometre give the corresponding route density. Pecuniary variables comprise GNP per capita and revenue per passenger kilometre travelled (or per freight tonne kilometre) as measures of the prices (fares or freight rates) faced by rail users. The price of petrol (or diesel) is also included. Finally we define two quality variables: frequency and station spacing. Frequency is measured as passenger (or freight) train kilometres

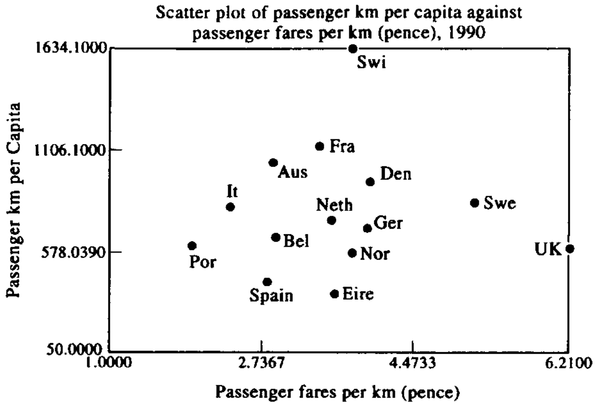

Figure 3

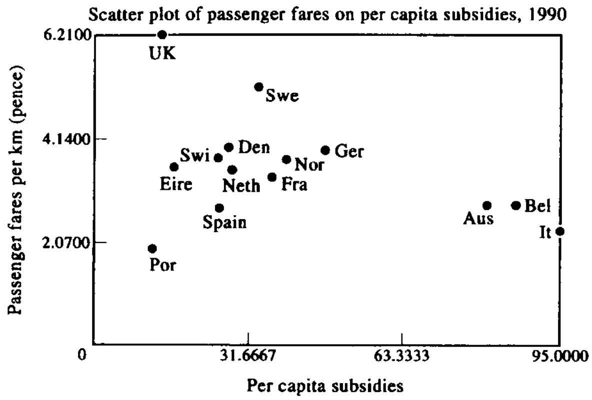

Figure 4

deflated by route length. Station spacing proxies the distance between stops though this, of course, is only a crude measure of actual points of access which was not directly available.

The tables also include the two dependent variables used in the regression analysis. Our measures of demand are passenger rail-km per capita and freight tonne-km per capita. As detailed in the introduction to the paper, the more general variable, modal share, which is the ratio of rail-km per capita to total-km per capita in terms of passengers or freight tonnes, is also available. However, its denominator contains estimated-km for other private modes which are subject to considerable measurement error, so this variable is less reliable than rail demand alone. Specifying modal share as the left-hand side variable also yielded inferior results so it was dropped from the econometric analysis.

Empirical specification

Guided by the available data and the elementary theory of consumer demand for intermediate services such as rail transport, which are purchased to move people and freight over desired routes, we proceed with the following specification, normalized in per capita terms:

$$D = F(\underset{+}{Y}, \underset{-}{TC}, \underset{+}{P})$$

Demand (D) is written down as a function of disposable income per head (Y), which we proxy by GNP per capita, the total cost of rail travel per unit distance (TC), and the price of the closest substitute (P). The expected signs of the partial derivatives are indicated beneath each argument in the function. Data constraints restrict our price of substitutes variable to the price of petrol or diesel as proxies for the marginal cost of private road transport per unit distance for passengers or freight respectively.

Our main innovation lies in the extended definition of total cost made possible by the new data set used here. The most commonly specified cost component of public transport is the average fare per passenger-km or per freight tonne-km. Government policy has pushed British passenger rail fares up to the highest level in Europe by a wide margin. This is illustrated in Figure 3 which plots passenger-km per capita against fare levels. Not unrelatedly, the UK has the highest private car transport share and the third lowest share for passenger rail in Europe (Preston *et al.*, 1994).

Even more fundamental than pecuniary price, the new microeconomics of consumer choice suggests that transport demand equations which lack careful quality controls or non-pecuniary costs are seriously mis-specified, for the opportunity cost of travel time is a, if not the, major cost component in transport modal choices. The door-to-door convenience of the private car or lorry represents a substantial time saving compared with the transfer and waiting times involved in station-to-station rail transportation. The comparative disadvantage of the rail mode does decline with the length of the trip, but remains substantial for most journeys in densely populated countries with good road networks.

We do not have data on the average speed of trains, which is inversely related to travel time, but we do have a number of proxies for the complementary time costs of travel from journey origin to connecting train station and the inconvenience cost of waiting time to which survey responses ascribe a high disutility. Perhaps the most obvious geographical variable that captures some of these complementary time costs is that of route density, that is railway route-km per square km. No less important in principle is the distance between stops, though this is unavailable. As a proxy we use station spacing, the distance between stations, which may be an unreliable indicator of actual train stops.

Waiting time is generally reduced by the frequency of service which we measure at the quotient of total train-km deflated by route length. Finally, as noted above, the relative time advantage of door to door road transport increases as the average journey length declines. This is most obvious for very short road trips, which may involve large detours by rail. As a proxy for average journey length we use population density, which is essentially a demographic-geographic constraint, in contrast to all other variables which are short- or long-run policy parameters.

In addition to the various cost components, there are also quality variables such as comfort, maintenance and

Table 3 OLS estimates for the rail passenger demand equations, 1990

Dependent variable	Passenger km per capita (1)		(2)	
Intercept	−14.99	(−2.74)	−14.17	(−3.69)
Population density	−0.69	(−2.03)	−0.76	(−2.45)
Route density	0.73	(1.83)	0.83	(2.26)
GNP	0.83	(1.45)	0.59	(1.32)
Price (passenger)	−0.10	(−0.23)	−0.44	(−1.57)
Frequency (passenger)	0.44	(1.54)	0.52	(2.16)
Petrol price	0.57	(0.88)		
Station spacing (passenger)	−0.20	(−0.81)		
$\bar{R}^2$	0.560		0.605	
$\hat{\sigma}$	0.259		0.245	
$\chi^2_H(1)$	0.812		0.566	

Notes: (i) t-statistics are in parentheses; (ii) all variables are in natural logarithms; (iii) $\bar{R}^2$ is the multiple correlation coefficient and $\hat{\sigma}$ is the estimated standard error of the regression; (iv) $\chi^2_H(1)$ is a diagnostic statistic distributed approximately as a chi-squared variate with one degree of freedom for a test of heteroscedasticity. For details of this diagnostic statistic, see Pesaran and Pesaran (1989); (v) prices are defined as passenger revenue per passenger-km; (iv) frequency is measured as passenger train-km deflated by route length.

Table 4 OLS estimates for the rail freight demand equations, 1990

Dependent variable	Freight tonne km per capita (1)		(2)	
Intercept	−19.90	(−3.28)	−13.82	(−3.55)
Population density	−1.22	(−5.38)	−1.33	(−5.64)
Route density	1.38	(2.88)	1.84	(4.27)
GNP	0.83	(1.22)	0.17	(0.28)
Price (freight)	−0.41	(−1.68)	−0.35	(−1.31)
Frequency (freight)	0.72	(2.54)	0.61	(2.23)
Diesel price	0.93	(1.55)		
Station spacing (freight)	−0.13	(−1.16)		
$\bar{R}^2$	0.884		0.859	
$\hat{\sigma}$	0.300		0.331	
$\chi^2_H(1)$	0.002		0.008	

Notes: (i) t-statistics are in parentheses; (ii) all variables are in natural logarithms; (iii) $\bar{R}^2$ is the multiple correlation coefficient and $\hat{\sigma}$ is the estimated standard error of the regression; (iv) $\chi^2_H(1)$ is a diagnostic statistic distributed approximately as a chi-squared variate with one degree of freedom for a test of heteroscedasticity. For details of this diagnostic statistic, see Pesaran and Pesaran (1989); (v) prices are defined as freight revenue per freight tonne-km; (iv) frequency is measured as freight train-km deflated by route length.

age of rolling stock which are likely to influence demand. We do not have direct proxies for these variables, but the general level of subsidies is available as a possible partial proxy for all aspects of quality including the time cost variables discussed above. However, for a given revenue, subsidies may also reflect inefficiences of rail management and operations. We do find a negative simple correlation (−0.24) between subsidies and fares showing that subsidies help to keep fares or freight rates low, but the subsidy variable did not perform well in (unreported) regressions. Figure 4 provides a plot of this inverse association. Note that only Portugal provides a smaller per capita level of direct government support for the rail system than the UK. Excluding Portugal, which represents an outlier in the sample, the negative correlation rises to −0.68.

One important variable which is omitted from the study, and which varies substantially across countries, is the degree of competition that rail, especially passenger rail, faces from alternative public modes of transport such as bus, coach and air. Inter-modal competition might be expected to affect the volume of rail demand though we have no readily available data to capture its impact.

Results

Log-linear demand equations for passenger rail and freight rail were estimated by Ordinary Least Squares using observations for fourteen countries. Given that the sample size is very small, the results that follow must be interpreted very cautiously. The sample size limits the number of degrees of freedom and the possibility of using more general functional forms, including interaction terms, that Wardman (1994) and others have used. Note also that we are unable to correct for possible simultaneity between rail demand and some of the explanatory variables. As Wardman points out, for example, a pure cross-sectional model may suffer from simultaneity bias insofar as, say, low frequency or a high fare level is the result of low rail demand rather than the cause of it. Wardman comments that such models tend to yield large elasticities for price and service quality relative to other models. In our equations, however, population density, and the route density and station spacing quality variables are not vulnerable to this critique since they are exogenously determined, at least in the medium run.

Tables 3 and 4 list the OLS estimates (t-ratios in parentheses) for passenger rail and freight rail respectively. Given the log-linear functional form, these estimates can be interpreted as elasticities. The results for the unrestricted specifications are reported in the first column of each table. The statistical performance of both equations is good for such a small sample. Heteroscedasticity cannot be detected and the goodness of fit is satisfactory for cross-section regressions. The signs of all of the estimated coefficients are consistent with prior expectations and have reasonable magnitudes. Given six degrees of freedom, the absolute critical value for statistical signficance at the 5% level is 1.94 (or 1.44 at the 10% level).

All but one of the parameters in the passenger equation are in fact insignificant at the 5% level. The exception is population density which is inversely related to the demand for rail, reflecting the comparative time advantage of private road transport for trips which will on average be of shorter duration in a densely populated country. We can simplify the equation by imposing zero restrictions on the coefficients of petrol price and station spacing in the passenger equation. Their t-ratios are less than unity (in absolute terms) and do not add to the R-bar-square. The restricted estimates are listed in column (2). The quality variables, route density and frequency, are now all statistically significant at the 5% level and price is significant at the 10% level. Only GNP remains insignificant at conventional levels.[5]

The freight equation is similar in many respects to the passenger rail regression, though the magnitudes of the coefficients tend to be larger, especially with respect to population and route densities. This suggests that the volume of freight rail is much more sensitive to changes in the explanatory variables than passenger demand. In both freight specifications, population density and the frequency and route density quality variables are well-determined, displaying the expected signs and statistical significance at conventional levels. Freight rates and diesel prices are only significant at the 10% level. The cross price elasticity of 0.93 on the diesel price does, however, provide some support to the proposal of the Royal Commission on Environmental Pollution (1994) to raise fuel duty on road freight as a powerful mechanism for inducing a modal switch in favour of rail.

Conclusion

In this paper we used new data on passenger and freight rail transport from fourteen European countries to explain the variation in international passenger and freight rail demand. We present OLS estimates of log-linear rail demand functions which include proxies for various non-pecuniary cost components namely: frequency; route density and population density variables. These non-pecuniary or quality factors in particular are demonstrated to have a significant role in explaining the cross-section variation in the rail demand data. While the frequency variable is particularly liable to simultaneity bias, its theoretical role in demand is unambiguous, so that the absence of this explanatory variable in many studies is likely to cause omitted variable bias.

With respect to passenger rail, Switzerland, the Netherlands and Eire were the most successful in maintaining their share of the market during the 1980s. This may be partly explained by the fact that the former two countries, in particular, are notable for their extremely high train frequencies. With regard to the levels and changes in the demand for freight rail, Switzerland and Austria are strong performers, again partly reflecting unusually high service frequencies.

The data also reveal in striking fashion how British rail transport has performed worse than almost all other European rail systems. Modal shares for both freight and passengers have declined faster between 1980 and 1990, and that from a lower level, than in many other European countries. In the passenger market, British Rail fares are the highest in the sample by a wide margin; in the freight market, the UK rail network has a remarkably large station spacing and, again, comparatively high haulage prices.

Moreover, British neglect of investment in infrastructure, including electrification and rolling stock, follows a pattern of government policy bias in favour of road transport that contrasts markedly with some of our European neighbours. The social costs resulting from these short-run savings in rail subsidies and investment are thus correspondingly greater in the UK. The political problem is that these costs are long term, widely diffused, and often hard to attribute unambiguously, thus blunting political pressure for their containment. The unresponsive nature of the UK political system exacerbates the problems, though as the recent Royal Commission Report makes clear, their magnitude is now so great that public concern is beginning to grow more outspoken.

[5] We also tested car ownership in unreported regressions, but this variable did not perform well either.

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